



An adaptive rank-based extension of Gaslike Social Motility for high-dimensional optimization

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CONTENT

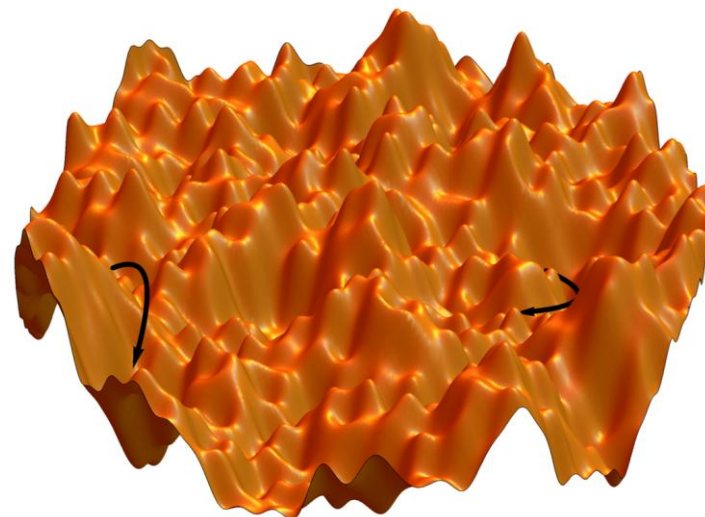
- 1.Introduction and motivation.
- 2.Problem statement and objective.
- 3.Original Gas-like Social Motility (GSM).
- 4.Proposed Method: GSM-R.
- 5.Experimental design.
- 6.Experimental results.
- 7.Conclusions and future work.

1. INTRODUCTION AND MOTIVATION

No Free Lunch Theorem

The High-Dimensional Challenge

Optimization in high-dimensional spaces remains highly challenging for evolutionary and swarm-based methods.



2. PROBLEM STATEMENT AND OBJECTIVE

Problem statement

- High-dimensional optimization increases search complexity.
- Fixed-radius neighborhoods become less informative.
- Raw fitness-based interactions become unstable.
- GSM performance degrades as dimensionality increases.

Objective

- Develop an adaptive extension of GSM.
- Improve exploration–exploitation balance.
- Increase robustness in high-dimensional spaces.

3. Original Gas-like Social Motility (GSM)

Algorithm Framework

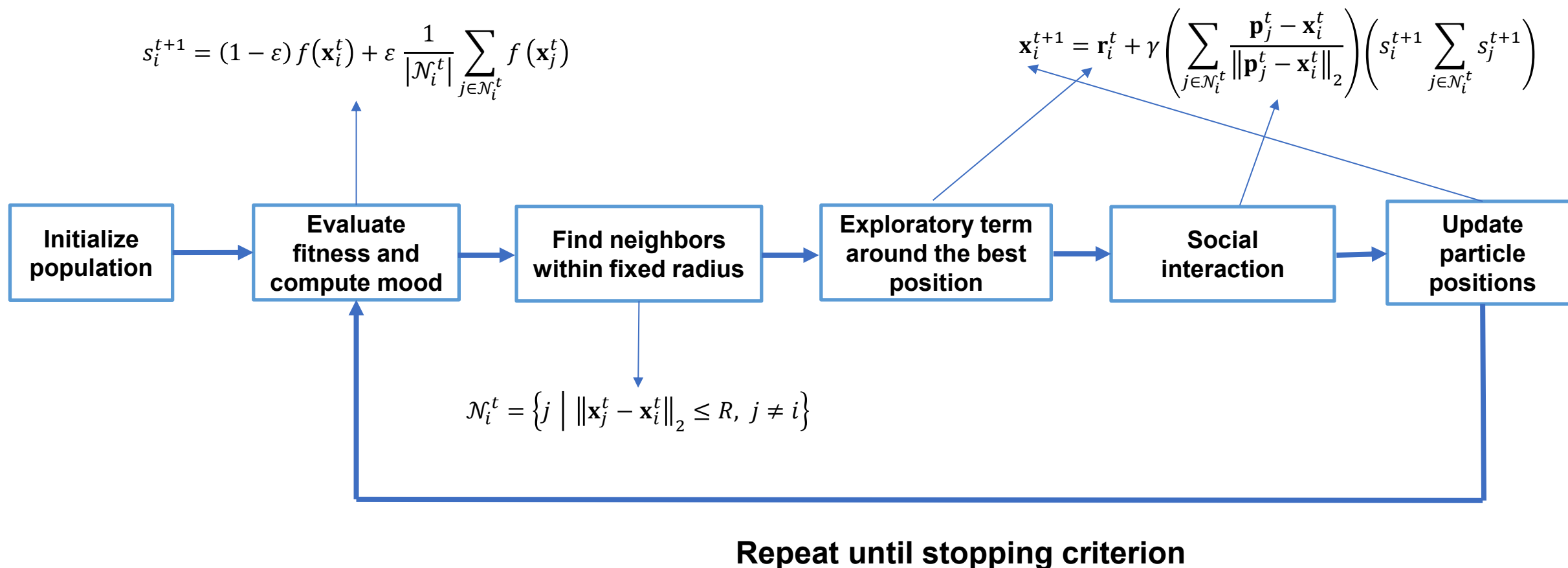
GSM is a population-based optimizer inspired by the collective movement of gas-like particles exhibiting social behavior.

- **Neighborhood:** Fixed interaction radius.
- **Movement:** The new position combines:
 - Exploration
 - Social movement

Critical Limitations

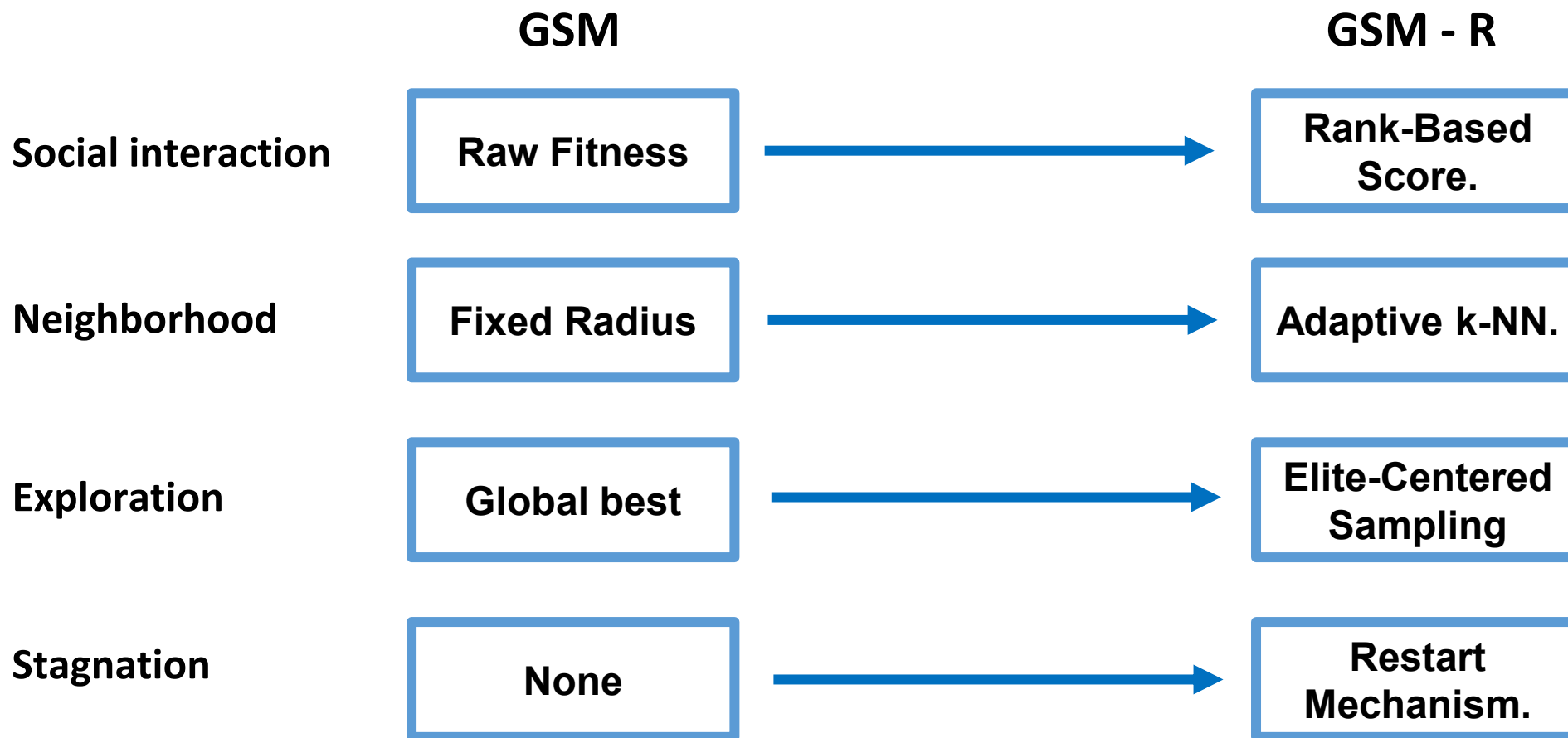
In highly complex spaces, distance-based fixed neighborhoods become obsolete. The raw fitness interaction causes severe scaling sensitivity, weakening the collective search process.

3. Original Gas-like Social Motility (GSM)



4. PROPOSED METHOD: GSM-R

Four Key Modifications



4. PROPOSED METHOD: GSM-R

Rank-Based Social Interaction

- **Addressing Fitness Scaling Issues:** At each iteration, the entire population is sorted according to objective value. The rank of particle i is then normalized into a score:

$$q_i^t = \left(1 - \frac{\text{rank}_i^t}{N - 1}\right)^\alpha, \quad \alpha > 0,$$

$$\tilde{q}_i^t = (1 - \varepsilon)q_i^t + \varepsilon \frac{1}{|\mathcal{N}_i^t|} \sum_{j \in \mathcal{N}_i^t} q_j^t,$$

$\text{rank}_i^t \in \{0, 1, \dots, N - 1\},$
 $\text{rank}_i^t = 0$ for the best particle.

4. PROPOSED METHOD: GSM-R

Social Direction and Interaction Gain

$$\mathbf{s}_i^t = \sum_{j \in \mathcal{N}_i^t} \frac{\mathbf{p}_j^t - \mathbf{r}_i^t}{\|\mathbf{p}_j^t - \mathbf{r}_i^t\|_2}$$

Social direction: combines the directions toward the personal best positions of neighboring particles.

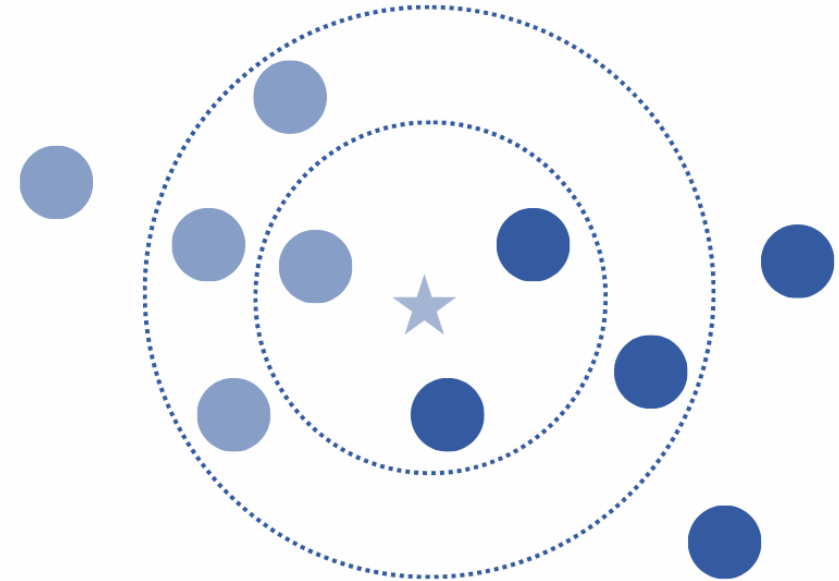
$$\beta_i^t = \text{clip} \left(\tilde{q}_i^t \sum_{j \in \mathcal{N}_i^t} \tilde{q}_j^t, 0, \beta_{\max} \right)$$

Interaction gain: controls the strength of the social movement and keeps it bounded.

4. PROPOSED METHOD: GSM-R

Adaptive k-Nearest Neighbor Interaction

- **Structural Shift:** The fixed interaction radius is entirely replaced. Instead, GSM-R defines social interaction through an adaptive k-nearest-neighbor (k-NN) structure.
- **Dynamic Adaptation:** The neighborhood size $k(t)$ dynamically shrinks as iterations progress. This creates a shift from global exploration (large k) to local exploitation (small k).



$$k(t) = \max \left(k_{\min}, \left\lceil \text{frac} N \left(1.05 - 0.25 \frac{t}{T-1} \right) \right\rceil \right)$$

4. PROPOSED METHOD: GSM-R

Elite-Centered Sampling

Instead of relying solely on the global best, GSM-R uses an elite subset to define a promising search region.

$$\mathbf{r}_i^{t+1} = \Pi_{[\ell,u]}^D \left(\underbrace{\mathbf{c}^t}_{\text{Elite region}} + \underbrace{\boldsymbol{\sigma}_{\text{eff}}^t \odot \mathbf{z}_i^t}_{\text{Exploration}} + \underbrace{\gamma \beta_i^t \mathbf{s}_i^t}_{\text{Social Movement}} \right)$$

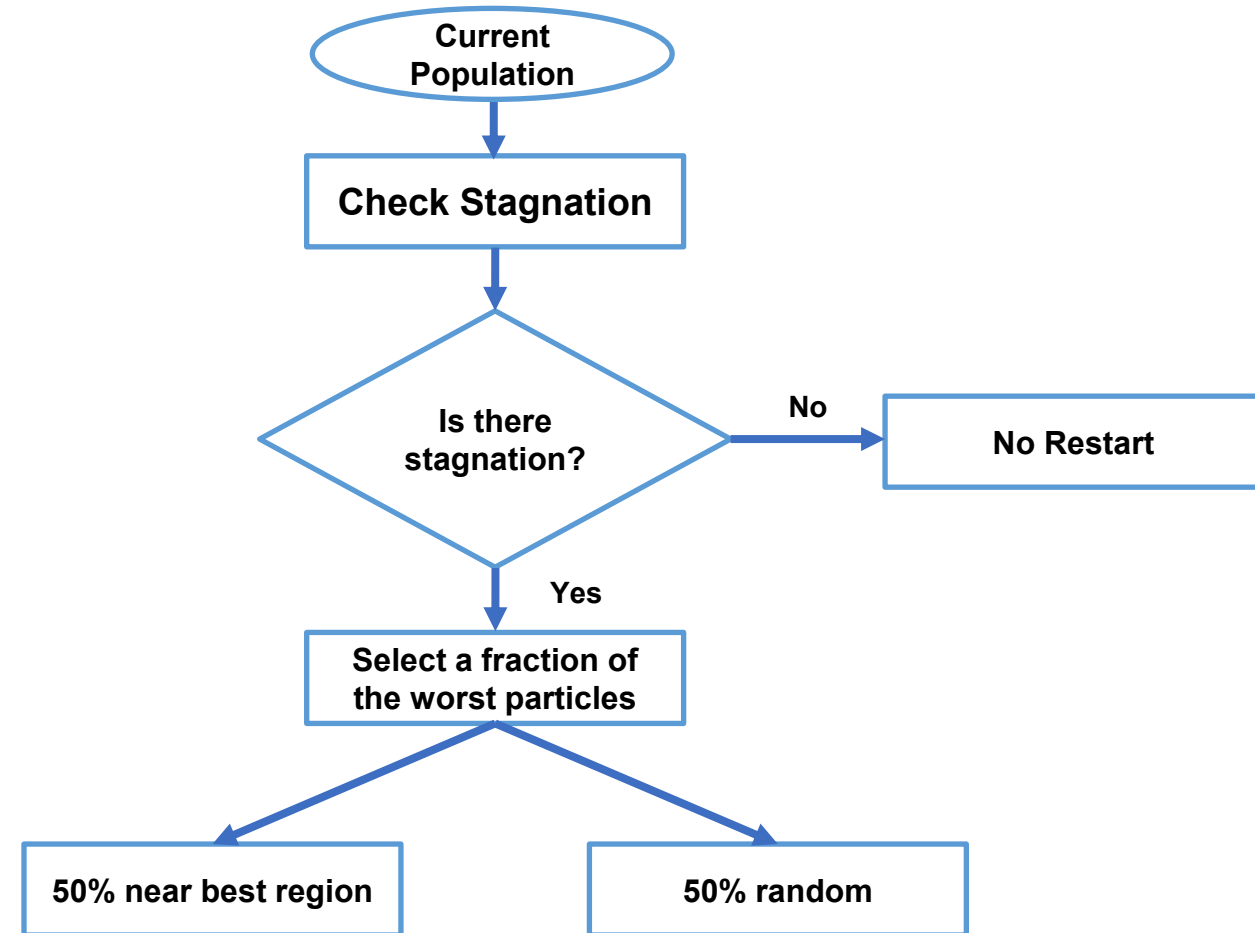
Diagram illustrating the components of the proposed method's position update equation:

- $\mathbf{r}_i^{t+1}$ is the **New Position**.
- $\Pi_{[\ell,u]}^D$ is the projection operator.
- $\mathbf{c}^t$ represents the **Elite region**.
- $\boldsymbol{\sigma}_{\text{eff}}^t \odot \mathbf{z}_i^t$ represents **Exploration**.
- $\gamma \beta_i^t \mathbf{s}_i^t$ represents **Social Movement**.

4. PROPOSED METHOD: GSM-R

Partial Restart

If the global best does not improve for a fixed number of iterations, a fraction of the worst particles is reinitialized. This mechanism is intended to mitigate prolonged stagnation while preserving the main population structure.



5. EXPERIMENTAL DESIGN

- **Benchmark Functions:** Evaluation conducted on **22 continuous functions** encompassing unimodal, multimodal, deceptive, non-separable, rugged, ill-conditioned, and shifted-rotated landscapes.
- **Dimensionality:** Testing performed on complex dimensionalities: $D = 50$ and $D = 100$, pushing the limits of standard optimization algorithms.
- **Compared Algorithms:** Benchmarked against standard **GSM, DE, PSO, GA, ES, LSHADE** and **SepCMAES**.
- **Configuration:** Population size of **40**, **35 iterations**, and **30 independent runs**. Designed specifically to simulate scenarios requiring rapid, early-stage optimization.

6. EXPERIMENTAL RESULTS

Table A.8: Median objective values obtained by all algorithms for $D = 50$. Lower values are better; the best result in each row is shown in bold.

Function	GSM	GSM-R	DE	PSO	GA	ES	LSHADE	SepCMAES
Rosenbrock	6.178e+05	3.050e+03	1.321e+06	4.784e+05	1.757e+06	1.677e+06	7.920e+03	1.253e+06
Rastrigin	5.169e+02	1.739e+02	6.667e+02	6.067e+02	5.416e+02	7.321e+02	3.260e+02	4.772e+02
Ackley	7.492e+00	2.197e+00	8.114e+00	7.048e+00	8.199e+00	9.079e+00	3.173e+00	4.078e+00
Schwefel	1.149e+04	1.147e+04	1.570e+04	1.297e+04	1.273e+04	1.651e+04	1.132e+04	1.725e+04
Levy	1.424e+02	8.725e+00	3.355e+02	1.598e+02	1.927e+02	4.629e+02	1.424e+01	3.018e+01
Weierstrass	5.690e+01	6.167e+01	7.705e+01	6.376e+01	7.280e+01	8.747e+01	2.994e+01	4.645e+01
Katsuura	6.485e+00	1.172e+02	2.520e-01	0.000e+00	2.181e+02	2.634e+01	3.560e+02	7.180e-01
Schwefel12	1.347e+05	2.887e+04	1.762e+05	1.230e+05	1.456e+05	1.917e+05	2.190e+04	1.285e+05
ExpScafferF6	2.298e+01	2.197e+01	2.368e+01	2.314e+01	2.240e+01	2.356e+01	2.134e+01	2.282e+01
LunacekBiRastrigin	5.840e+02	2.260e+02	8.665e+02	6.511e+02	6.496e+02	8.159e+02	3.982e+02	7.372e+02
SumDiffPowers	1.456e-01	8.986e-05	5.005e-01	4.809e-02	7.003e-02	1.193e+00	1.100e-07	1.340e-03
Schwefel221	8.367e+01	6.362e+01	9.411e+01	7.765e+01	8.446e+01	9.138e+01	3.448e+01	4.915e+01
Ridge	2.784e+10	1.843e+09	7.572e+10	3.911e+10	7.805e+10	1.138e+11	2.211e+09	8.350e+09
Schwefel223	3.819e+02	1.908e-02	9.737e+02	6.982e+01	3.099e+02	1.621e+03	1.881e-02	1.002e-01
Griewank	4.050e+02	3.391e+01	8.258e+02	4.097e+02	7.445e+02	1.004e+03	2.072e+01	7.275e+01
Elliptic	2.052e+06	7.257e+04	4.151e+06	1.469e+06	4.523e+06	4.115e+06	6.034e+04	1.561e+06
Discus	2.013e+02	5.520e+00	2.801e+02	4.033e+02	4.309e+02	5.690e+02	5.138e+01	3.800e+02
BentCigar	1.127e+08	2.821e+06	2.137e+08	7.671e+07	2.105e+08	2.858e+08	5.123e+06	1.647e+07
GriewankRosenbrock	1.846e+05	1.184e+03	2.276e+06	9.585e+04	4.855e+05	2.036e+06	2.866e+02	9.148e+02
ShiftedRotRastrigin	5.070e+02	4.034e+02	7.484e+02	6.495e+02	6.571e+02	7.512e+02	4.156e+02	5.848e+02
ShiftedRotAckley	6.552e+00	3.236e+00	8.405e+00	7.216e+00	9.347e+00	8.911e+00	3.509e+00	8.963e+00
ShiftedRotElliptic	1.435e+06	5.172e+05	5.236e+06	1.700e+06	6.084e+06	4.983e+06	2.286e+05	4.902e+06

6. EXPERIMENTAL RESULTS

Table A.9: Median objective values obtained by all algorithms for $D = 100$. Lower values are better; the best result in each row is shown in bold.

Function	GSM	GSM-R	DE	PSO	GA	ES	LSHADE	SepCMAES
Rosenbrock	4.287e+06	2.006e+05	4.832e+06	2.303e+06	5.077e+06	6.895e+06	2.267e+05	3.039e+06
Rastrigin	1.307e+03	5.815e+02	1.437e+03	1.361e+03	1.282e+03	1.600e+03	8.668e+02	1.075e+03
Ackley	8.357e+00	3.995e+00	8.657e+00	8.388e+00	9.128e+00	9.715e+00	4.193e+00	5.651e+00
Schwefel	2.638e+04	2.743e+04	3.492e+04	3.006e+04	3.005e+04	3.590e+04	2.789e+04	3.650e+04
Levy	4.402e+02	5.443e+01	8.835e+02	5.846e+02	5.868e+02	1.076e+03	8.443e+01	2.306e+02
Weierstrass	1.436e+02	1.194e+02	1.653e+02	1.580e+02	1.657e+02	1.873e+02	8.109e+01	1.200e+02
Katsuura	3.835e+01	4.643e+02	2.203e-01	0.000e+00	1.595e+03	1.214e+02	2.272e+03	4.461e+01
Schwefel12	3.944e+05	1.420e+05	6.081e+05	4.462e+05	6.020e+05	6.472e+05	8.088e+04	6.022e+05
ExpScafferF6	4.700e+01	4.617e+01	4.818e+01	4.794e+01	4.658e+01	4.806e+01	4.468e+01	4.703e+01
LunacekBiRastrigin	1.530e+03	7.039e+02	2.035e+03	1.482e+03	1.673e+03	2.044e+03	1.028e+03	1.433e+03
SumDiffPowers	4.248e-01	1.345e-04	5.326e-01	7.471e-01	1.594e-01	1.836e+00	6.364e-07	3.965e-03
Schwefel221	9.180e+01	6.756e+01	9.110e+01	9.783e+01	9.084e+01	9.637e+01	4.575e+01	9.760e+01
Ridge	1.785e+11	1.636e+10	1.997e+11	1.585e+11	2.288e+11	2.563e+11	1.758e+10	3.842e+10
Schwefel223	2.343e+03	4.900e+00	1.846e+03	1.845e+03	2.212e+03	6.128e+03	9.063e-01	3.730e+01
Griewank	1.532e+03	1.224e+02	1.791e+03	1.518e+03	1.941e+03	2.490e+03	1.407e+02	4.249e+02
Elliptic	7.525e+06	2.965e+05	1.683e+07	6.693e+06	2.249e+07	1.999e+07	4.651e+05	1.068e+07
Discus	6.987e+02	5.167e+01	5.352e+02	9.419e+02	7.343e+02	9.822e+02	1.624e+02	6.702e+02
BentCigar	4.282e+08	3.934e+07	4.170e+08	3.886e+08	5.602e+08	6.525e+08	5.231e+07	1.241e+08
GriewankRosenbrock	3.697e+06	6.207e+03	4.545e+06	1.574e+06	3.447e+06	8.906e+06	2.201e+03	9.527e+04
ShiftedRotRastrigin	1.504e+03	7.784e+02	1.517e+03	1.538e+03	1.435e+03	1.691e+03	1.058e+03	1.318e+03
ShiftedRotAckley	9.015e+00	4.383e+00	9.494e+00	8.940e+00	1.010e+01	1.012e+01	5.413e+00	8.968e+00
ShiftedRotElliptic	7.395e+06	1.275e+06	1.760e+07	6.543e+06	1.977e+07	1.780e+07	1.195e+06	1.496e+07

6. EXPERIMENTAL RESULTS

Table 2: Overall performance comparison across all benchmark functions and both dimensionalities.

Algorithm	Wins	Avg. Rank
GSM-R	22	1.77
LSHADE	19	1.86
PSO	2	4.48
GSM	1	4.36
SepCMAES	0	3.93
GA	0	5.95
DE	0	6.18
ES	0	7.45

Table 3: Number of wins obtained by each algorithm in each dimensionality.

Algorithm	$D = 50$	$D = 100$
LSHADE	11	8
GSM-R	10	12
PSO	1	1
GSM	0	1
SepCMAES	0	0
GA	0	0
DE	0	0
ES	0	0

Table 4: Average rank obtained by each algorithm for each dimensionality. Lower values indicate better performance.

Algorithm	$D = 50$	$D = 100$
LSHADE	1.77	1.95
GSM-R	1.91	1.64
SepCMAES	3.86	4.00
GSM	4.18	4.55
PSO	4.32	4.64
GA	6.09	5.82
DE	6.55	5.82
ES	7.32	7.59

Table 5: Wilcoxon signed-rank test comparing GSM-R against competing algorithms.

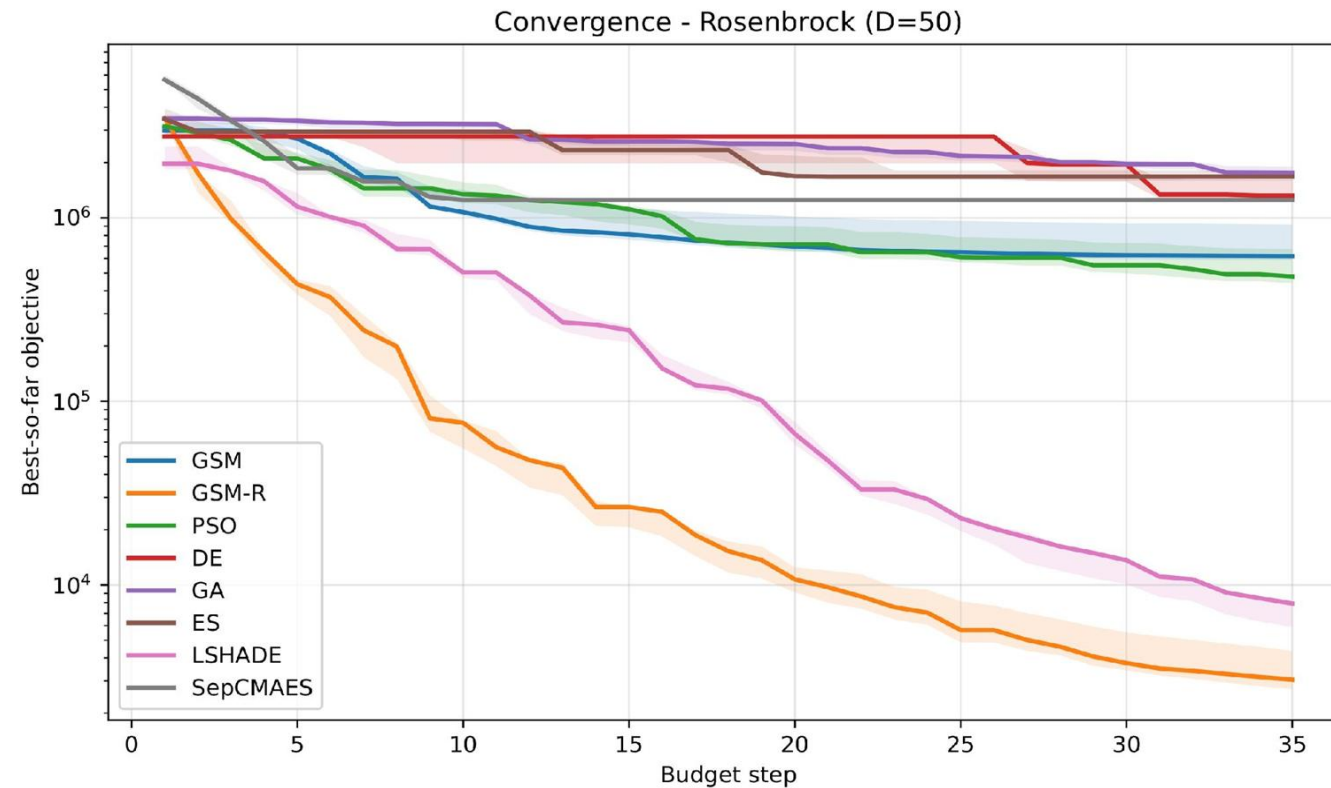
Comparison	$D = 50$		$D = 100$	
	p -value	Sig.	p -value	Sig.
GSM-R vs ES	$4.53e-06$	Yes	$4.53e-06$	Yes
GSM-R vs GA	$2.38e-07$	Yes	$2.38e-07$	Yes
GSM-R vs DE	$4.53e-06$	Yes	$4.53e-06$	Yes
GSM-R vs PSO	$5.96e-06$	Yes	$4.53e-06$	Yes
GSM-R vs GSM	$2.10e-05$	Yes	$1.07e-04$	Yes
GSM-R vs LSHADE	$4.12e-01$	No	$6.44e-02$	No
GSM-R vs SepCMAES	$9.59e-04$	Yes	$1.03e-05$	Yes

Table 6: Friedman test results and average ranks for all algorithms in each dimensionality.

Dim.	χ^2	p -value	DE	ES	GA	GSM	GSM-R	LSHADE	PSO	SepCMAES
50	107.61	$2.88e-20$	6.55	7.32	6.09	4.18	1.91	1.77	4.32	3.86
100	102.59	$3.14e-19$	5.82	7.59	5.82	4.55	1.64	1.95	4.64	4.00

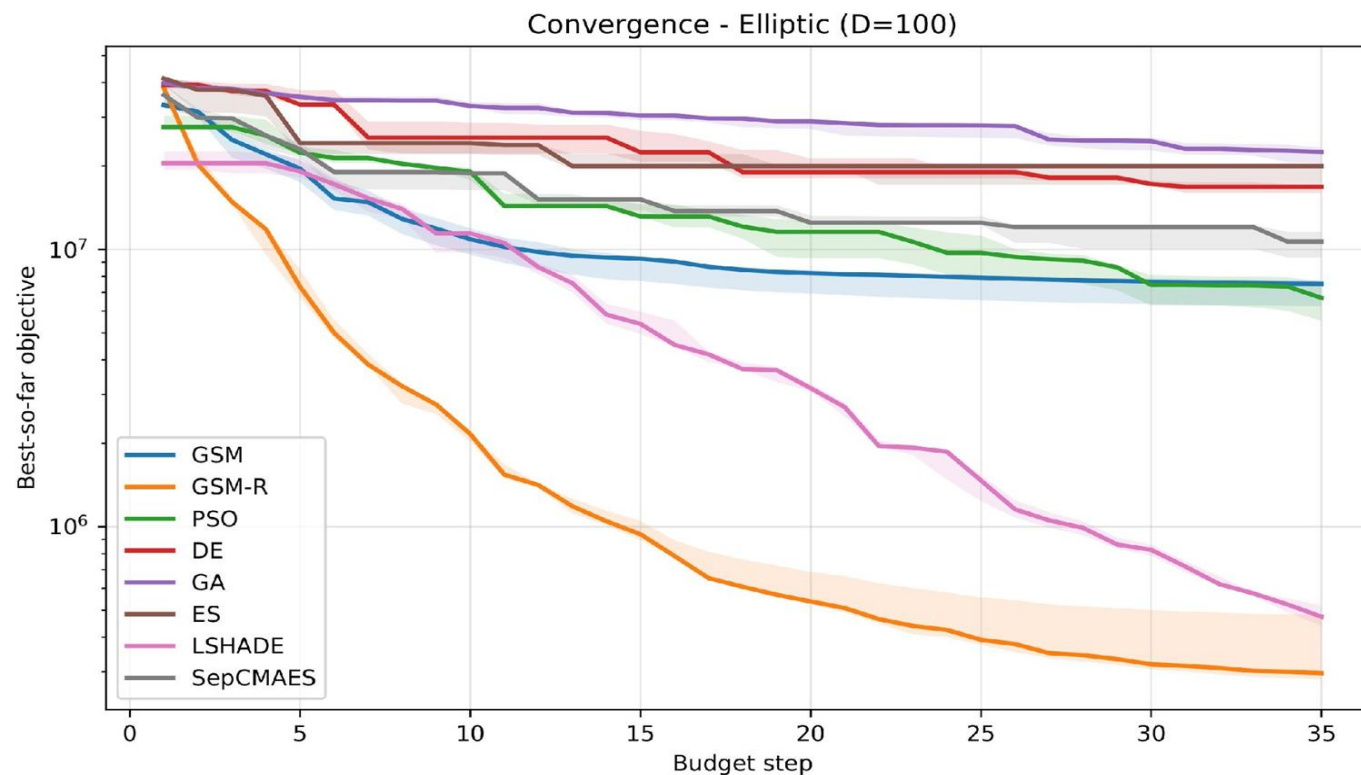
6. EXPERIMENTAL RESULTS

Convergence Curves



6. EXPERIMENTAL RESULTS

Convergence Curves



6. EXPERIMENTAL RESULTS

Ablation Study

Table 7: Ablation study: Median objective values on a subset of $D = 100$ functions. Best results are in bold.

Function	GSM (Original)	GSM-R (w/o Restart)	GSM-R (w/o Elite)	GSM-R (Full)
Rosenbrock	4.287e+06	1.767e+05	3.097e+06	1.702e+05
Rastrigin	1.307e+03	5.556e+02	1.293e+03	5.462e+02
Ridge	1.785e+11	1.589e+10	1.713e+11	1.774e+10
ShiftedRotElliptic	7.395e+06	1.195e+06	3.852e+06	1.147e+06

6. CONCLUSIONS AND FUTURE WORK

Overall Performance Results

22 Total Benchmark wins

GSM-R achieved the highest total number of wins across the full benchmark set (out of 44 total dimensional tests), beating all baseline algorithms.

Furthermore, it achieved an overall **average rank of 1.77**, proving high consistency. In the highly complex $D = 100$ setting, GSM-R secured 12 direct wins compared to 8 for its closest rival, LSHADE.

Statistically, Wilcoxon signed-rank and Friedman tests confirmed GSM-R's superiority over standard methods and statistical equivalence to the highly adaptive LSHADE framework.



Thank you!

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